ThreatLens AI

DIGITAL RISK PROTECTION & BRAND DEFENSE PLATFORM

DOCUMENTATION SPECIFICATION
Platform Version: 2.0.0 PRO • October 2026
Evaluation & Architecture Reference

Executive Summary: ThreatLens AI is a next-generation Digital Risk Protection (DRP) and cybersecurity intelligence platform designed to safeguard enterprise brands against lookalike phishing domains, executive social impersonation, rogue mobile applications, and visual brand theft. Registered official assets bypass threat scoring with guaranteed zero false positives, while suspicious indicators are scored deterministically using explainable heuristics and vision OCR forensics.

1. Platform REST APIs Overview (Next.js App Router)

ThreatLens exposes 8 high-performance serverless REST API endpoints running inside Next.js with automatic database connection fallback:

API Route	Methods	Functionality & Purpose	Key Parameters / Payload
/api/analyze	POST	**URL & Multi-Channel Threat Scanner:** Evaluates submitted URLs, social handles, or app package IDs. Executes SSRF guards, checks authoritative brand baseline exclusions, and calculates 0–100 risk score with plain-English threat explanations.	`candidate, channel (domain/social/app), platform, bio, publisher, saveToFindings`
/api/analyze-screenshot NEW	POST	**Screenshot Vision & OCR Forensics:** Ingests uploaded screenshot images, extracts embedded URLs, profile handles, and bio lures via OCR, detects visual spoofing cues (fake badges, cloned forms), and returns interactive hotspots with a full threat intelligence report.	`image (Data URL), sampleId, customCandidate, customChannel, saveToFindings`

/api/findings	GET
	POST
	PATCH
		Threat Telemetry & Findings Database: Fetches, filters, ingests, and updates threat findings. Allows SOC analysts to filter by risk level (Critical/High/Medium/Low) or status, and update triage status (new, reviewed, confirmed, false_positive).	`GET: ?type=&platform=&riskLevel=&status=&search=`
			`PATCH: { id, status }`
			`POST: { type, displayName, riskScore, signals }`

API Route	Methods	Functionality & Purpose	Key Parameters / Payload

/api/brand	GET POST DELETE	**Authoritative Brand Baseline:** Manages enterprise verified assets (registered domains, official social handles, verified mobile bundle IDs, VIP executive roster). Guarantees zero false positives on legitimate assets.	`GET: returns brand baseline`
			`POST: { type, platform, value }`
			`DELETE: ?id=<assetId>`

/api/investigations	GET POST	**SOC Incident Case Management:** Manages formalized investigation cases. Allows analysts to bundle multiple threat findings into active incident tickets, associate cryptographic evidence hashes, and track audit notes.	`GET: list cases with evidence & notes` `POST: { title, description, severity, findingIds }`
/api/audit-logs	GET	**Security Audit Trail:** Provides an immutable timeline log of all analyst actions, asset additions, automated scans, and triage modifications for compliance standards (SOC2 Type II / NIST CSF).	`take=50, ordered by timestamp descending`
/api/export	GET	**Compliance & Threat Audit Export:** Downloads a comprehensive, structured JSON report containing all monitored telemetry, baseline assets, investigations, metrics, and forensic audit trails.	`Content-Disposition: attachment; filename="ThreatLens-Audit-Export.json"`
/api/reset	POST	**Factory Demo Baseline Reset:** Wipes and restores the database to an authoritative demo baseline with realistic threat scenarios, verified assets, sample investigations, and integration states.	`POST: triggers demo re-seeding`

2. External Integrations & Intelligence APIs

VirusTotal Intelligence API v3

OpenAI GPT-4o / AI Copilot

Endpoint: `https://www.virustotal.com/api/v3`

Cross-references detected domain hosts, IPs, and uploaded files against global reputation blacklists, AV engine detections, and historical malicious passive DNS telemetry.

Model: `gpt-4o-mini / gpt-4o`

Synthesizes transparent heuristic results into plain-English threat narratives, summarizes complex forensic indicators, and drafts legally compliant DMCA / registrar takedown notices.

3. AI Vision & Screenshot Forensic Scanner Details

The newly integrated **Screenshot Impersonation & Phishing Scanner** allows users and analysts to identify threats from images without manually retyping obscure URLs or handles:

- **Multi-Format Ingestion:** Drag-and-drop or file upload for PNG, JPG, JPEG, and WEBP. Supports direct system clipboard paste (**Ctrl + V**) after capturing a screenshot.
- **Optical Character Recognition (OCR) Engine:** Extracts address bar URLs (including deceptive TLDs like `.xyz`, `.co`), social media handles (`@threatlens_support_bot`), and Android package names (`com.threatlens.guard.mod`).
- **Visual Spoofing Detection:** Identifies cloned corporate Single Sign-On (SSO) login forms, counterfeit brand vector marks, emoji checkmarks used to fake verified status, and predatory permission requests.
- **Interactive Hotspot Pins:** Displays numbered interactive pins directly over the screenshot image highlighting the exact location of detected fraud (Address Bar, Profile Bio, Download Link).
- **1-Click Remediation:** Instant export of formal legal Takedown Notices for domain registrars, hosting providers, or social platform trust & safety teams.

4. Complete Platform Technology Stack

Layer	Technologies Used	Role in ThreatLens Architecture
Frontend Framework	Next.js 16 (App Router) + React 19	Server components, client interactivity, and streaming dashboard views.
Styling & UI System	Vanilla Tailwind CSS 4	Modern light-theme SaaS design (#F7F9FC canvas, white cards, soft borders).
Type Safety	TypeScript 5.x	Strict end-to-end typing across database models, APIs, and heuristic payloads.
Database & ORM	Prisma ORM 5.22 + SQLite	Local persistent database with graceful in-memory fallback for serverless hosting.
Heuristic Security Engine	Deterministic Rules (heuristics.ts)	100% offline-capable, explainable scoring: Homoglyphs, Protocol Backslash, Combisquatting.
Iconography	Lucide React	High-clarity SVG security badges, radar indicators, and threat severity icons.
Deployment Targets	Netlify & Vercel	Automated CI/CD with pinned Node 20 runtime and serverless functions.